

AVONDALE 2 N, PA, USA

WMO: 725160

Lat: 39.8593N

Lon: 75.7861W

Elev: 122

StdP: 99.87

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03761

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.3	-21.0	0.6	-12.2	-18.4	0.7	-9.9	7.6	-1.9	7.1	-2.1	2.3	N/A	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.6	23.8	30.3	23.0	29.1	22.3	25.1	29.6	24.2	28.5	23.5	27.4	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.8	27.6	22.9	17.9	26.4	22.3	17.3	25.7	76.8	29.6	73.5	28.6	70.6	27.5	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.7	4.8	-15.7	34.0	2.9	1.7	-17.7	35.2	-19.4	36.2	-21.0	37.2	-23.1	38.4		
(5)			-16.8	26.5	2.7	0.6	-18.7	27.0	-20.3	27.4	-21.8	27.7	-23.7	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-0.9	0.2	4.8	10.9	16.6	20.9	23.4	22.1	18.9	12.8	6.5	2.5		
	DBStd	9.52	5.64	5.59	5.36	4.82	4.23	3.09	2.59	2.62	3.67	4.63	4.48	4.98		
	HDD10.0	1224	339	277	177	44	2	0	0	0	0	24	123	239		
	HDD18.3	2911	595	507	419	227	86	12	1	2	38	180	354	491		
	CDD10.0	1818	2	4	16	72	206	326	415	376	266	111	18	7		
	CDD18.3	464	0	0	0	5	32	87	157	120	54	9	0	0		
	CDH23.3	3642	0	0	3	61	270	699	1312	880	367	51	0	0		
	CDH26.7	1009	0	0	0	10	62	182	430	225	93	7	0	0		
(14) Wind	WSAvg	1.6	2.1	2.3	2.2	2.0	1.3	1.3	1.0	1.0	1.1	1.4	1.7	1.9		
(15) Precipitation	PrecAvg	1191	94	65	113	97	88	114	104	102	123	102	86	100		
	PrecMax	1634	155	147	194	187	179	287	199	196	326	264	127	205		
	PrecMin	881	41	12	12	42	36	48	29	41	10	18	26	24		
	PrecStd	209	34	28	43	39	37	66	46	39	81	64	31	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	19.7	23.0	28.3	30.6	32.8	33.9	32.3	31.5	27.2	20.7	18.5		
		MCWB	14.1	14.0	15.7	18.8	22.0	23.8	25.6	25.0	23.2	20.5	15.4	16.4		
	2%	DB	13.4	14.9	19.3	25.1	28.5	30.4	31.9	30.5	29.4	24.6	18.4	15.5		
		MCWB	11.2	11.4	13.7	17.4	21.0	22.3	24.2	23.4	22.2	20.1	14.6	14.2		
	5%	DB	10.2	11.7	16.1	22.4	26.4	28.8	30.5	29.1	27.3	22.5	16.1	12.9		
		MCWB	7.5	7.9	11.8	16.3	19.9	21.6	23.6	22.4	21.2	18.4	12.2	11.1		
	10%	DB	7.5	8.3	13.4	19.9	24.4	27.2	29.1	27.8	25.3	20.4	14.1	10.1		
		MCWB	5.2	5.3	9.1	14.3	18.5	20.7	22.8	21.9	20.1	16.9	10.8	7.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.7	16.7	20.2	23.5	24.8	26.7	25.7	24.7	22.5	17.7	16.9		
		MCDB	15.3	19.0	21.4	24.1	28.9	30.3	32.1	30.0	29.1	24.8	19.4	18.0		
	2%	WB	10.8	12.2	14.6	18.7	22.2	23.6	25.5	24.7	23.7	20.9	15.5	14.9		
		MCDB	12.3	14.0	18.2	22.8	26.9	28.4	30.4	28.5	27.3	23.6	17.6	15.8		
	5%	WB	7.8	8.7	12.2	16.8	21.1	22.8	24.6	23.8	22.8	19.6	13.5	12.3		
		MCDB	9.5	11.6	15.8	21.4	25.4	26.8	29.1	27.3	25.7	22.1	15.5	13.5		
	10%	WB	5.6	6.0	9.7	15.1	20.0	21.9	23.8	22.8	21.8	17.8	11.5	9.2		
		MCDB	7.4	8.8	13.2	18.6	23.6	25.3	27.9	25.8	24.2	20.2	13.6	10.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.7	9.7	11.0	10.6	10.4	10.1	10.0	10.3	10.2	9.5	8.0		
		MCDBR	11.7	12.9	13.9	14.2	12.8	12.0	11.3	11.0	12.2	12.0	12.1	10.5		
		MCWBR	10.2	9.5	9.6	7.6	6.4	5.5	4.7	4.9	5.5	6.8	8.8	8.7		
		MCDBR	11.1	12.5	13.0	12.3	11.4	10.1	9.8	9.2	9.3	9.3	10.4	9.4		
	5% WB	MCWBR	10.1	10.1	9.5	7.9	6.2	5.3	4.7	4.6	4.9	6.4	8.9	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.318	0.356	0.397	0.439	0.503	0.508	0.506	0.418	0.358	0.323	0.306		
	taud		2.451	2.409	2.375	2.280	2.196	2.055	2.057	2.049	2.269	2.444	2.508	2.503		
	Ebn at Noon		858	901	898	878	844	788	780	768	821	843	837	835		
	Edh at Noon		80	95	110	129	144	166	164	161	122	92	75	71		
(44) All-Sky Solar Radiation	RadAvg		1.92	2.82	3.72	4.82	5.39	5.92	5.98	5.24	4.31	3.10	2.21	1.67		
	RadStd		0.16	0.32	0.29	0.42	0.65	0.46	0.28	0.32	0.50	0.33	0.19	0.17		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+74

Nomenclature: See separate page